Functions: Composite, Inverse & Logarithmic

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domain and range**

Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

The denominator is zero when $x = \pm 3$. Domain: all real x except $x = 3$ and $x = -3$.

2 **[Foundational]** • **Target: Domain and range**

Find the domain of $g(x) = \sqrt{7 - 2x}$.

The radicand must be non-negative: $7 - 2x \geq 0 \Rightarrow x \leq \frac{7}{2}$.

3 **[Foundational]** • **Target: Domain and range**

Find the range of $f(x) = -(x - 2)^2 + 5$.

Since $-(x - 2)^2 \leq 0$ for all x, $f(x) \leq 5$. Range: $f(x) \leq 5$.

4 **[Foundational]** • **Target: Composite functions**

Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $(f \circ g)(x)$.

$f(g(x)) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10$.

5 **[Foundational]** • **Target: Composite functions**

Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(x)$.

$f(g(x)) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = |x|$.

6 **[Foundational]** • **Target: Composite functions**

Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.

$g(f(x)) = \left(\sqrt{x + 4}\right)^2 - 4 = (x + 4) - 4 = x$, valid for $x \geq -4$ (the domain of f).

7 **[Foundational]** • **Target: Inverse functions**

Find $f^{-1}(x)$ for $f(x) = \frac{2x - 5}{3}$.

$y = \frac{2x - 5}{3} \Rightarrow 3y = 2x - 5 \Rightarrow x = \frac{3y + 5}{2}$. So $f^{-1}(x) = \frac{3x + 5}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Inverse functions

Find $f^{-1}(x)$ for $f(x) = x^3 + 2$.

$y = x^3 + 2 \Rightarrow x = \sqrt[3]{y - 2}$. So $f^{-1}(x) = \sqrt[3]{x - 2}$.

9 [Intermediate] • Target: Verifying inverses

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses by showing $f(g(x)) = x$.

$f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x - 1) + 1 = x$. Confirmed.

10 [Intermediate] • Target: Verifying inverses

Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.

No. f is not one-to-one — it fails the horizontal line test, since for example $f(2) = f(-2) = 4$. An inverse function only exists if the domain is restricted (e.g. to $x \geq 0$).

11 [Intermediate] • Target: Logarithm laws

Expand $\log_b\left(\frac{x^3 y}{z^2}\right)$ using the laws of logarithms.

$\log_b\left(\frac{x^3 y}{z^2}\right) = 3\log_b x + \log_b y - 2\log_b z$.

12 [Intermediate] • Target: Logarithm laws

Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

$2\log_5 x + 3\log_5 y - \log_5 z = \log_5\left(\frac{x^2 y^3}{z}\right)$.

13 [Intermediate] • Target: Logarithm laws

Evaluate $\log_2 32 - \log_2 4$ without a calculator.

$\log_2\left(\frac{32}{4}\right) = \log_2 8 = 3$.

14 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_4(x^2) = 3$.

$x^2 = 4^3 = 64 \Rightarrow x = \pm 8$. Both satisfy the domain requirement $x^2 > 0$. Solutions: $x = 8$ or $x = -8$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Solve $\ln x + \ln(x - 3) = \ln 10$.

$\ln[x(x - 3)] = \ln 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0 \Rightarrow x = 5$ or $x = -2$. Since $\ln(x - 3)$ requires $x > 3$, reject $x = -2$. Solution: $x = 5$.

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16 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $e^{2x-1} = 20$, giving your answer to three decimal places.

$2x - 1 = \ln 20 \approx 2.996 \Rightarrow x = \frac{1 + 2.996}{2} \approx 1.998$.

17 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 5.01 Trigonometry

The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

$81 = 3^4$, so $x = 4$. Since f and g are inverses, $g(f(16)) = 16$.

18 [Challenging] • Target: Transformations of functions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The graph shows $f(x) = x^2$ (teal) and $g(x) = (x-2)^2 + 1$ (indigo). Describe the transformation that maps f onto g.

g is f shifted right 2 units and up 1 unit — a horizontal translation of $+2$ followed by a vertical translation of $+1$.

19 [Challenging] • Target: Transformations of functions • Cross-Lesson: 7.01 Solid Geometry & Similarity

If $h(x) = -f(x+3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h.

Shift f left 3 units (from $x + 3$), then reflect the result over the x-axis (from the negative sign outside).

20 [Challenging] • Target: A well-designed word problem: investment growth and doubling time • Cross-Lesson: 8.01 Statistics & Probability

This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t, to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.

(a) $A(10) = 8000e^{0.45} \approx 8000(1.5683) \approx 12546$ dollars. (b) Doubling requires $e^{0.045t} = 2 \Rightarrow 0.045t = \ln 2 \Rightarrow t = \frac{\ln 2}{0.045} \approx 15.4$ years. (c) From $A = A_0 e^{rt}$: $\frac{A}{A_0} = e^{rt} \Rightarrow \ln\left(\frac{A}{A_0}\right) = rt \Rightarrow t(A) = \frac{1}{0.045}\ln\left(\frac{A}{A_0}\right)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Domain and range

If $f(x) = -4x + 7$ and $g(x) = -2x - 5$, find $(f \circ g)(7)$.

a. 83 ✓

b. 85

c. -19

d. 81

$g(7) = -2(7) - 5 = -19$. $f(-19) = -4(-19) + 7 = 83$.

22 [Foundational] • Target: Composite functions

If $f(x) = x$ and $g(x) = -3x + 3$, find $(f \circ g)(6)$.

a. -13

b. -15 ✓

c. -12

d. -17

$g(6) = -3(6) + 3 = -15$. $f(-15) = -15$.

23 [Foundational] • Target: Inverse functions

If $f(x) = 2x - 6$ and $g(x) = -3x + 1$, find $(f \circ g)(3)$.

a. -20

b. -22 ✓

c. -8

d. -24

$g(3) = -3(3) + 1 = -8$. $f(-8) = 2(-8) - 6 = -22$.

24 [Foundational] • Target: Verifying inverses

If $f(x) = x$ and $g(x) = 3x - 9$, find $(f \circ g)(4)$.

a. 6

b. 5

c. 1

d. 3 ✓

$g(4) = 3(4) - 9 = 3$. $f(3) = 3$.

25 [Foundational] • Target: Logarithm laws

If $f(x) = x - 1$ and $g(x) = 2x + 4$, find $(f \circ g)(-4)$.

a. -3

b. -7

c. -4

d. -5 ✓

$g(-4) = 2(-4) + 4 = -4$. $f(-4) = (-4) - 1 = -5$.

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26 [Foundational] • Target: Solving logarithmic equations

If $f(x) = 4x + 8$ and $g(x) = 2x - 2$, find $(f \circ g)(-7)$.

a. -56 ✓

b. -16

c. -58

d. -54

$g(-7) = 2(-7) - 2 = -16$. $f(-16) = 4(-16) + 8 = -56$.

27 [Foundational] • Target: Exponential equations and logs as inverses

If $f(x) = -4x + 5$ and $g(x) = 2x + 3$, find $(f \circ g)(-2)$.

a. 11

b. 7

c. -1

d. 9 ✓

$g(-2) = 2(-2) + 3 = -1$. $f(-1) = -4(-1) + 5 = 9$.

28 [Foundational] • Target: Transformations of functions

Find $f^{-1}(x)$ for $f(x) = \frac{6x-5}{8}$.

a. $\frac{6x+5}{8}$

b. $\frac{8x+5}{6}$ ✓

c. $\frac{8x-5}{6}$

d. $\frac{6x-5}{8}$

$y = \frac{6x-5}{8} \Rightarrow 8y = 6x - 5 \Rightarrow x = \frac{8y+5}{6}$. So $f^{-1}(x) = \frac{8x+5}{6}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: investment growth and doubling time

Find $f^{-1}(x)$ for $f(x) = \frac{9x-13}{4}$.

a. $\frac{9x-13}{4}$

b. $\frac{4x-13}{9}$

c. $\frac{9x+13}{4}$

d. $\frac{4x+13}{9}$ ✓

$y = \frac{9x-13}{4} \Rightarrow 4y = 9x - 13 \Rightarrow x = \frac{4y+13}{9}$. So $f^{-1}(x) = \frac{4x+13}{9}$.

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30 [Intermediate] • Target: Inverses & Domain Restrictions

Find $f^{-1}(x)$ for $f(x) = \frac{5x-14}{3}$.

a. $\frac{5x-14}{3}$

b. $\frac{3x-14}{5}$

c. $\frac{3x+14}{5}$ ✓

d. $\frac{5x+14}{3}$

$y = \frac{5x-14}{3} \Rightarrow 3y = 5x - 14 \Rightarrow x = \frac{3y+14}{5}$. So $f^{-1}(x) = \frac{3x+14}{5}$.

31 [Intermediate] • Target: Composition & Decomposition of Functions

Find $f^{-1}(x)$ for $f(x) = \frac{7x-15}{4}$.

a. $\frac{4x+15}{7}$ ✓

b. $\frac{7x-15}{4}$

c. $\frac{7x+15}{4}$

d. $\frac{4x-15}{7}$

$y = \frac{7x-15}{4} \Rightarrow 4y = 7x - 15 \Rightarrow x = \frac{4y+15}{7}$. So $f^{-1}(x) = \frac{4x+15}{7}$.

32 [Intermediate] • Target: Logarithmic Laws & Base Changes

Find $f^{-1}(x)$ for $f(x) = \frac{5x-2}{8}$.

a. $\frac{5x-2}{8}$

b. $\frac{8x-2}{5}$

c. $\frac{5x+2}{8}$

d. $\frac{8x+2}{5}$ ✓

$y = \frac{5x-2}{8} \Rightarrow 8y = 5x - 2 \Rightarrow x = \frac{8y+2}{5}$. So $f^{-1}(x) = \frac{8x+2}{5}$.

33 [Intermediate] • Target: Exponential Models & Logarithmic Graphs

Find $f^{-1}(x)$ for $f(x) = \frac{2x-11}{6}$.

a. $\frac{2x-11}{6}$

b. $\frac{2x+11}{6}$

c. $\frac{6x-11}{2}$

d. $\frac{6x+11}{2}$ ✓

$y = \frac{2x-11}{6} \Rightarrow 6y = 2x - 11 \Rightarrow x = \frac{6y+11}{2}$. So $f^{-1}(x) = \frac{6x+11}{2}$.

34 [Intermediate] • Target: Domain and range

Find $f^{-1}(x)$ for $f(x) = \frac{6x-3}{3}$.

a. $\frac{6x-3}{3}$

b. $\frac{6x+3}{3}$

c. $\frac{3x-3}{6}$

d. $\frac{3x+3}{6}$ ✓

$y = \frac{6x-3}{3} \Rightarrow 3y = 6x - 3 \Rightarrow x = \frac{3y+3}{6}$. So $f^{-1}(x) = \frac{3x+3}{6}$.

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35 [Intermediate] • Target: Composite functions

Solve $\ln x + \ln(x-5) = \ln 6$.

a. 8

b. -1

c. 4

d. 6 ✓

$\ln[x(x-5)] = \ln 6 \Rightarrow x(x-5) = 6 \Rightarrow x^2 - 5x - 6 = 0 \Rightarrow (x-6)(x-(-1)) = 0 \Rightarrow x = 6$ (rejecting $x = -1$ since $x > 5$ is required).

36 [Intermediate] • Target: Inverse functions

Solve $\ln x + \ln(x-6) = \ln 247$.

a. -13

b. 17

c. 21

d. 19 ✓

$\ln[x(x-6)] = \ln 247 \Rightarrow x(x-6) = 247 \Rightarrow x^2 - 6x - 247 = 0 \Rightarrow (x-19)(x-(-13)) = 0 \Rightarrow x = 19$ (rejecting $x = -13$ since $x > 6$ is required).

37 [Intermediate] • Target: Verifying inverses

Solve $e^{3x-7} = 105$, giving your answer to three decimal places.

a. 3.385

b. 3.885 ✓

c. 4.885

d. 4.385

$3x - 7 = \ln 105 \approx 4.654 \Rightarrow x = \frac{7 + 4.654}{3} \approx 3.885$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Logarithm laws • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve $e^{4x-6} = 488$, giving your answer to three decimal places.

a. 4.048

b. 3.548

c. 3.048 ✓

d. 2.548

$4x - 6 = \ln 488 \approx 6.190 \Rightarrow x = \frac{6 + 6.190}{4} \approx 3.048$.

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39 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Evaluate $\log_5 500 - \log_5 4$ without a calculator.

a. 496

b. 3 ✓

c. 4

d. 2

$\log_5\left(\frac{500}{4}\right) = \log_5 125 = 3$.

40 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 8.01 Statistics & Probability

Evaluate $\log_3 72 - \log_3 8$ without a calculator.

a. 64

b. 2 ✓

c. 3

d. 1

$\log_3\left(\frac{72}{8}\right) = \log_3 9 = 2$.

41 [Challenging] • Target: Inverses & Domain Restrictions • Cross-Lesson: 1.01 Advanced Algebra

If $f(x) = \frac{2x-5}{3x+1}$, what is the value of $f^{-1}(1)$?

a. -6 ✓

b. -5

c. -4

d. -2

Set $f(x) = 1$: $\frac{2x-5}{3x+1} = 1 \Rightarrow 2x - 5 = 3x + 1 \Rightarrow -x = 6 \Rightarrow x = -6$.

42 [Challenging] • Target: Inverses & Domain Restrictions • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Given $f(x) = x^2 - 4x + 7$ defined on the restricted domain $[2, \infty)$, what is the equation of $f^{-1}(x)$?

a. $f^{-1}(x) = 2 + \sqrt{x-3}$ ✓

b. $f^{-1}(x) = 2 - \sqrt{x-3}$

c. $f^{-1}(x) = \sqrt{x-7} + 2$

d. $f^{-1}(x) = 2 + \sqrt{x+3}$

Complete the square: $y = (x-2)^2 + 3 \Rightarrow y - 3 = (x-2)^2$. For $x \geq 2$: $x - 2 = \sqrt{y-3} \Rightarrow x = 2 + \sqrt{y-3}$. Thus $f^{-1}(x) = 2 + \sqrt{x-3}$.

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43 [Challenging] • Target: Composition & Decomposition of Functions • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If $h(x) = (2x^2 - 1)^5$, identify inner function $g(x)$ and outer function $f(u)$ such that $h(x) = f(g(x))$.

a. $g(x) = 2x^2 - 1$ **and** $f(u) = u^5$ ✓

b. $g(x) = x^5$ and $f(u) = 2u^2 - 1$

c. $g(x) = 2x$ and $f(u) = (u - 1)^5$

d. $g(x) = 2x^2$ and $f(u) = u^5 - 1$

Direct decomposition gives inner $g(x) = 2x^2 - 1$ and outer power function $f(u) = u^5$.

44 [Challenging] • Target: Logarithmic Laws & Base Changes • Cross-Lesson: 5.01 Trigonometry

Using the change of base formula, evaluate $\log_8(32)$.

a. $\frac{5}{3}$ ✓

b. $\frac{3}{5}$

c. 4

d. $\frac{4}{3}$

Express with base 2: $\log_8(32) = \frac{\log_2 32}{\log_2 8} = \frac{5}{3}$.

45 [Challenging] • Target: Exponential Models & Logarithmic Graphs • Cross-Lesson: 6.01 Circles & Coordinate Geometry

What is the vertical asymptote of the graph of $f(x) = 3\ln(2x - 8) + 1$?

a. $x = 4$ ✓

b. $x = 8$

c. $x = 0$

d. $y = 1$

The argument of the natural log equals zero at the vertical asymptote: $2x - 8 = 0 \Rightarrow 2x = 8 \Rightarrow x = 4$.

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